Class 10: Structural Bioinformatics 1

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB).

Introduction to the RCSB Protein Data Bank (PDB)

The PDB archive is the major repository of information about the 3D structures of large biological molecules, including proteins and nucleic acids. Understanding the shape of these molecules helps to understand how they work. This knowledge can be used to help deduce a structure's role in human health and disease, and in drug development. The structures in the PDB range from tiny proteins and bits of DNA or RNA to complex molecular machines like the ribosome composed of many chains of protein and RNA.

In the first section of this lab we will interact with the main US based PDB website.

PDB Statistics and Data Import

What is in the PDB in terms of molecule type and structure determination method.

Read a CSV file of current PDB stats obtained from https://www.rcsb.org/stats/summary

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

```
           Molecular.Type   X.ray     EM    NMR Integrative Multiple.methods
1          Protein (only) 180,758 23,111 12,813         348              229
2 Protein/Oligosaccharide  10,488  3,741     34           8               11
3              Protein/NA   9,205  6,751    287          26                8
4     Nucleic acid (only)   3,154    250  1,578           3               15
5                   Other     178     27     35           4                0
6  Oligosaccharide (only)      11      0      6           0                1
  Neutron Other   Total
1      84    32 217,375
2       1     0  14,283
3       0     0  16,277
4       3     1   5,004
5       0     0     244
6       0     4      22
```

> **Q1**: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

The percentage of structures in the PDB solved by X-Ray is 80.48577%. The percentage of structures in the PDB that are solved by Electron Microscopy is 13.38046%.

```
pdb$X.ray
```

```
[1] "180,758" "10,488"  "9,205"   "3,154"   "178"     "11"
```

This print out adove `pdb$X.ray` is "character" not "numeric". Therefore I can't do math with it. We need to fix this...

Two functions that can help here are `sub()` and `as.numeric()`

```
# We want to get rid (or sub out) commas:
x <- pdb$X.ray
tmp <- sub(",", "", x)
sum(as.numeric(tmp))
```

```
[1] 203794
```

We could make a function to do this:

```r
rm.comma <- function(x) {
  tmp <- gsub(",", "", x)
  sum(as.numeric(tmp))
}
```

```r
n.tot <- rm.comma(pdb$Total)
n.xray <- rm.comma(pdb$X.ray)
n.em <- rm.comma(pdb$EM)

n.xray/n.tot *100
```

```
[1] 80.48577
```

```r
n.em/n.tot * 100
```

```
[1] 13.38046
```

We could also use a different import function 'for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file)

```r
library(readr)

read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# A tibble: 6 x 9
  `Molecular Type`    `X-ray`    EM   NMR Integrative `Multiple methods` Neutron
  <chr>                 <dbl> <dbl> <dbl>       <dbl>              <dbl>   <dbl>
1 Protein (only)       180758 23111 12813         348                229      84
2 Protein/Oligosacch~   10488  3741    34           8                 11       1
3 Protein/NA             9205  6751   287          26                  8       0
4 Nucleic acid (only)    3154   250  1578           3                 15       3
5 Other                   178    27    35           4                  0       0
6 Oligosaccharide (o~      11     0     6           0                  1       0
# i 2 more variables: Other <dbl>, Total <dbl>
```

Q2: What proportion of structures in the PDB are protein?

The proportion of structures in the PDB that are protein is 97.92%.

```
pdb$Total[1]
```

```
[1] "217,375"
```

The total number of protein sequences is 202,556,314, according to Uniprot database.

```
 (217375/202556314) *100
```

```
[1] 0.1073158
```

Key-point: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~%80) come from one method (X-ray crystalography)

```
# Find proportion of protein structures in the PDB database by using data from rows 1-3 and c
n.tot <- rm.comma(pdb$Total)
n.pro <- rm.comma(pdb$Total[1:3])

(n.pro / n.tot)
```

```
[1] 0.9791868
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 1227 HIV-1 protease structures in the current PDB.

Visualizing PDB data with Mol-star

Main stand alone web version with all features is at https://molstar.org/viewer/.

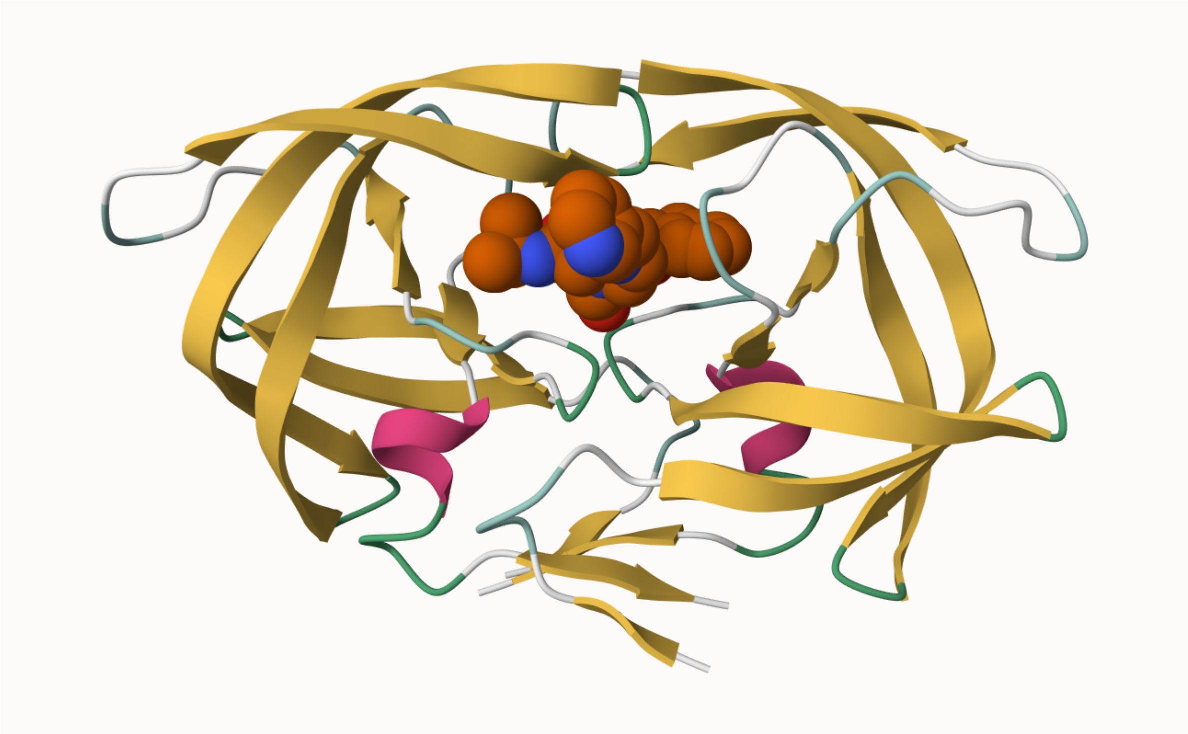


Figure 1: The HIV-protease enzyme is a homodimer with two chains

Figure 2: The HIV-protease enzyme is a homodimer with two chains

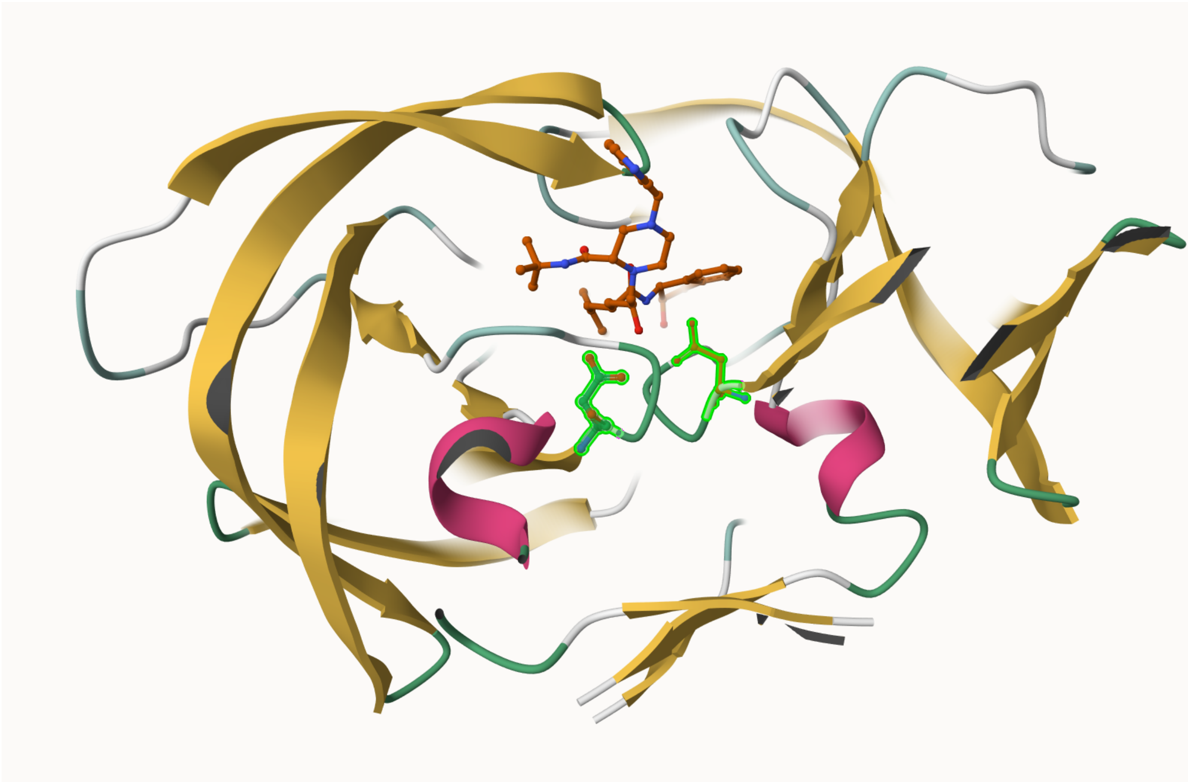


Figure 3: The HIV-protease enzyme with both Asp 25 positions selected

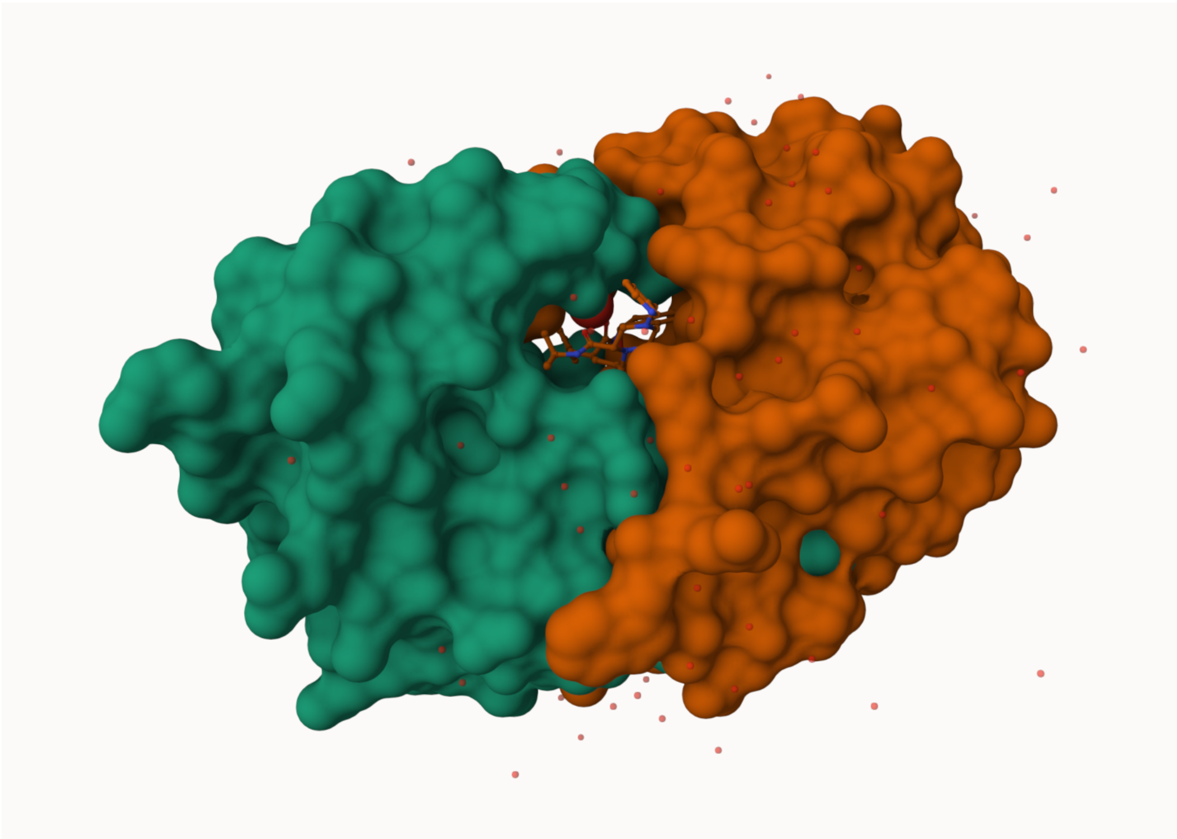


Figure 4: HIV-Pr molecular surface representation display of polymer showing the binding cleft site for the inhibitor (drug) molecule

> **Q4**: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Since hydrogen only has one electron, it is not visible with the methods used to solve/find these structures. Electron microscopy and X-ray diffraction better detect the oxygen molecule. Therefore, only one atom (oxygen) is used to represent the water molecules in this structure.

> **Q5**: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

This water molecule is HOH 308.

> **Q6**: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.

Figure 5: HIV-protease cartoon polymer with both chains A and B, the ligand, both Asp 25 residues selected, and the critical water molecule

```
# Pak can install from multiple sources
#install.packages("pak")

#| eval: !expr knitr::is_html_output()

pak::pkg_install(c("bioboot/bio3dview",
                   "NGLVieweR",
                   "bioc::msa"))
```

```
! Using bundled GitHub PAT. Please add your own PAT using `gitcreds::gitcreds_set()`.

  Loading metadata database

  Loading metadata database ... done
```

```
No downloads are needed

3 pkgs + 47 deps: kept 46 [6s]
```

```
#install.packages("bio3d")
library(bio3d)
```

Getting started with the Bio3D package

Bio3D is an R package from CRAN for structural bioinformatics

```
library(bio3d)

pdb <- read.pdb("1hsg")
pdb
```

> **Q7**: How many amino acid residues are there in this pdb object?

There are 198 amino acid residues in this pdb object.

> **Q8**: Name one of the two non-protein residues?

MK1 is one of the two non-protein residues.

> **Q9**: How many protein chains are in this structure?

There are 2 protein chains in this structure.

```
head(pdb$atom)
```

```
NULL
```

There are lots of functions that can work with these `pdb` objects:

```
#head( pdbseq(pdb))
```

```
#| eval: !expr knitr::is_html_output()

library(bio3dview)
library(NGLVieweR)

#| eval: !expr knitr::is_html_output()
#view.pdb(pdb) |> setSpin()
```

```
#install.packages("remotes")
remotes::install_github("bioboot/bio3dview")
```

```
Skipping install of 'bio3dview' from a github remote, the SHA1 (568c6e99) has not changed si
  Use `force = TRUE` to force installation
```

```
#install.packages("NGLVieweR")
```

let's try a custom view

```
#view.pdb(pdb,colorScheme="sse", backgroundColor="black")
```

> Q. Create a custom view highlighting the active site ASP (`resno25`), the two chains (in your choice of colors) and the ligand all on a custom color backgroud

```
library(NGLVieweR)
active.site <- atom.select(pdb, resno=25)

#| eval: !expr knitr::is_html_output()

#view.pdb(pdb,
         cols= c("red", "blue"),
         highlight = active.site,
         highlight.style = "spacefill",
         backgroundColor= "pink") |>
  setRock()
```

Predict the flexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given `pdb` object:

```r
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

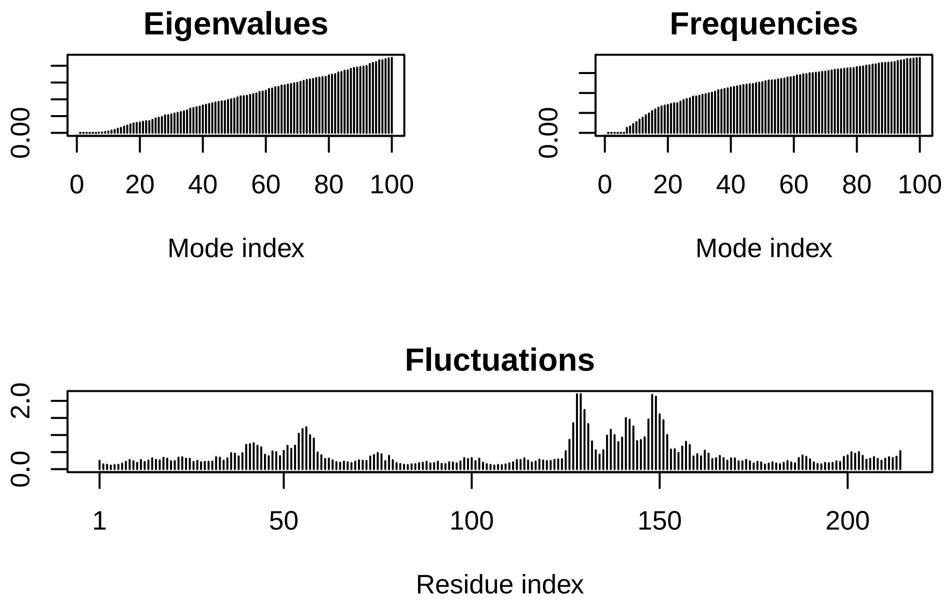
```r
adk
```

```
 Call:  read.pdb(file = "6s36")

   Total Models#: 1
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
m <- nma( adk )
```

```
 Building Hessian...        Done in 0.012 seconds.
 Diagonalizing Hessian...   Done in 0.282 seconds.
```

```r
plot(m)
```



View the results with an interactive structure view

```
view.nma(m)
```

Write out the results for viewing in Mol-star:

```
mktrj(m, file="nma.pdb")
```

Comparative analysis of the ADK family

> **Q10**. Which of the packages above is found only on BioConductor and not CRAN?

msa is found only on BioConductor.

> **Q11**. Which of the above packages is not found on BioConductor or CRAN?

bio3dview is not found on Bioconductor or CRAN.

> **Q12**. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

This is true.

Our first step is find a sequence for this family. We will use the database ID “1ake_A” here:

```r
id <- "1ake_A"

aa <- get.seq(id)
```

```
Warning in get.seq(id): Removing existing file: seqs.fasta
```

```
Fetching... Please wait. Done.
```

```r
aa
```

```
               1        .         .         .         .         .         60
pdb|1AKE|A    MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
               1        .         .         .         .         .         60

              61        .         .         .         .         .         120
pdb|1AKE|A    DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
              61        .         .         .         .         .         120

             121        .         .         .         .         .         180
pdb|1AKE|A    VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
             121        .         .         .         .         .         180

             181        .         .         .   214
pdb|1AKE|A    YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
             181        .         .         .   214

Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

There are 214 amino acids in this sequence.

Search for related sequences in the database

```r
blast <- blast.pdb(aa)
```

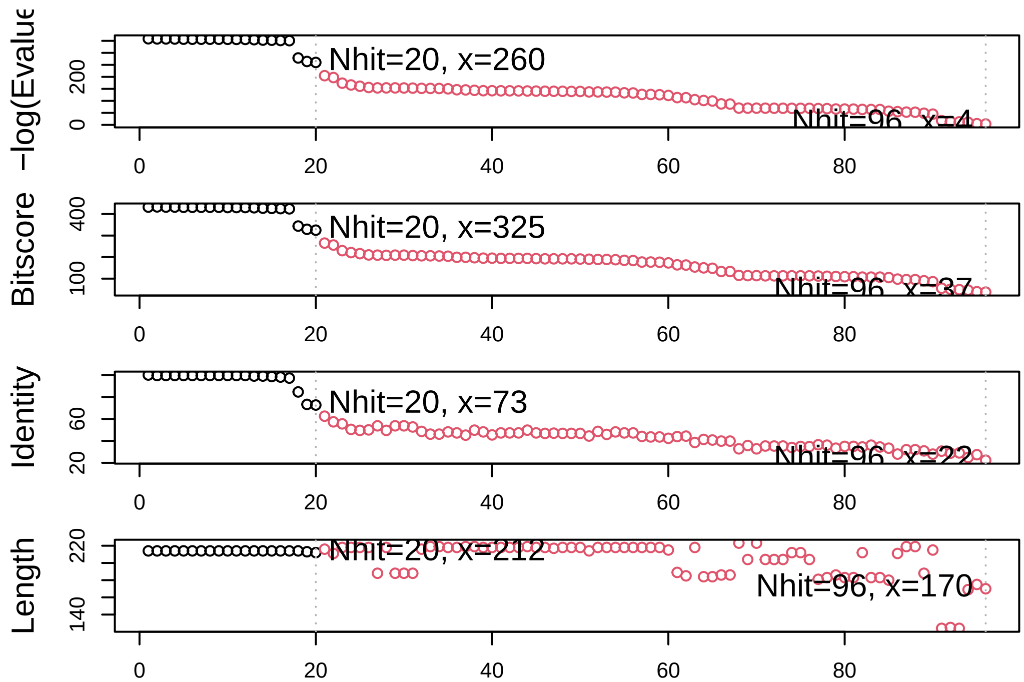
```
 Searching ... please wait (updates every 5 seconds) RID = 03XYGS5H016
 ....
 Reporting 96 hits
```

```r
head(blast$hit.tbl)
```

```
        queryid subjectids identity alignmentlength mismatches gapopens q.start
1 Query_5681485     1AKE_A  100.000             214          0        0       1
2 Query_5681485     8BQF_A   99.533             214          1        0       1
3 Query_5681485     4X8M_A   99.533             214          1        0       1
4 Query_5681485     6S36_A   99.533             214          1        0       1
5 Query_5681485     9R6U_A   99.533             214          1        0       1
6 Query_5681485     9R71_A   99.533             214          1        0       1
  q.end s.start s.end    evalue bitscore positives mlog.evalue pdb.id    acc
1   214       1   214 1.82e-156      432    100.00    358.6044 1AKE_A 1AKE_A
2   214      21   234 2.97e-156      433    100.00    358.1147 8BQF_A 8BQF_A
3   214       1   214 3.25e-156      432    100.00    358.0246 4X8M_A 4X8M_A
4   214       1   214 4.77e-156      432    100.00    357.6409 6S36_A 6S36_A
5   214       1   214 1.06e-155      431     99.53    356.8424 9R6U_A 9R6U_A
6   214       1   214 1.25e-155      431     99.53    356.6775 9R71_A 9R71_A
```

```r
hits <- plot(blast)
```

```
  * Possible cutoff values:    260 3
            Yielding Nhits:    20 96

  * Chosen cutoff value of:    260
            Yielding Nhits:    20
```




```
hits$pdb.id
```

```
 [1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
 [9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
# Download releated PDB files
files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download


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  |
  |======================================================================| 100%
```

```r
# Align releated PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/8BQF_A.pdb
pdbs/split_chain/4X8M_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/9R6U_A.pdb
pdbs/split_chain/9R71_A.pdb
pdbs/split_chain/8Q2B_A.pdb
pdbs/split_chain/8RJ9_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/4X8H_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/8PVW_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/4NP6_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
```

```
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
....   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

Extracting sequences

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/8BQF_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbs/split_chain/6S36_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/9R6U_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbs/split_chain/9R71_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/8Q2B_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbs/split_chain/8RJ9_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbs/split_chain/6RZE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10   name: pdbs/split_chain/4X8H_A.pdb
pdb/seq: 11   name: pdbs/split_chain/3HPR_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 13   name: pdbs/split_chain/5EJE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 15   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 16   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 17   name: pdbs/split_chain/6HAM_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18   name: pdbs/split_chain/8PVW_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19   name: pdbs/split_chain/4K46_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 20   name: pdbs/split_chain/4NP6_A.pdb
```

```
pdbs
```

```
                               1        .         .         .         40 
[Truncated_Name:1]1AKE_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:2]8BQF_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:3]4X8M_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:4]6S36_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:5]9R6U_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:6]9R71_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:7]8Q2B_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:8]8RJ9_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:9]6RZE_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:10]4X8H_A.pdb  --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:11]3HPR_A.pdb  --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:12]1E4V_A.pdb  --MRIILLGAPVAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:13]5EJE_A.pdb  --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:14]1E4Y_A.pdb  --MRIILLGALVAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:15]3X2S_A.pdb  --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:16]6HAP_A.pdb  --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:17]6HAM_A.pdb  --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:18]8PVW_A.pdb  --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:19]4K46_A.pdb  --MRIILLGAPGAGKGTQAQFIMAKFGIPQISTGDMLRAA
[Truncated_Name:20]4NP6_A.pdb  NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTGDMLRAA
                                 ********  *********** *^************** 
                               1        .         .         .         40 

                              41        .         .         .         80 
[Truncated_Name:1]1AKE_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:2]8BQF_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQE---
[Truncated_Name:3]4X8M_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:4]6S36_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:5]9R6U_A.pdb   VKSGSELGAQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:6]9R71_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:7]8Q2B_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:8]8RJ9_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:9]6RZE_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:10]4X8H_A.pdb  VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:11]3HPR_A.pdb  VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:12]1E4V_A.pdb  VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:13]5EJE_A.pdb  VKSGSELGKQAKDIMDACKLVTDELVIALVKERIAQEDCR
[Truncated_Name:14]1E4Y_A.pdb  VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:15]3X2S_A.pdb  VKSGSELGKQAKDIMDCGKLVTDELVIALVKERIAQEDSR
```

```
[Truncated_Name:16]6HAP_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVRERICQEDSR
[Truncated_Name:17]6HAM_A.pdb   IKSGSELGKQAKDIMDAGKLVTDEIIIALVKERICQEDSR
[Truncated_Name:18]8PVW_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:19]4K46_A.pdb   IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA
[Truncated_Name:20]4NP6_A.pdb   IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE
                                ^* *^*** *** ^^*   **^*^^^^^*^^*** *
                               41         .         .         .        80

                               81         .         .         .        120
[Truncated_Name:1]1AKE_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:2]8BQF_A.pdb    -GFLLDGFPRTIPQADAMKEAGINVDYVIEFDVPDELIVD
[Truncated_Name:3]4X8M_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:4]6S36_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:5]9R6U_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:6]9R71_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDALIVD
[Truncated_Name:7]8Q2B_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:8]8RJ9_A.pdb    NGFLLAGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:9]6RZE_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:10]4X8H_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:11]3HPR_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:12]1E4V_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:13]5EJE_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:14]1E4Y_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:15]3X2S_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:16]6HAP_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:17]6HAM_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:18]8PVW_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:19]4K46_A.pdb   KGFLLDGFPRTIPQADGLKEVGVVVDYVIEFDVADSVIVE
[Truncated_Name:20]4NP6_A.pdb   KGFLLDGFPRTIPQADGLKEMGINVDYVIEFDVADDVIVE
                                 **** **********^^** *^ ****^**** * ^**^
                               81         .         .         .        120

                              121         .         .         .        160
[Truncated_Name:1]1AKE_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:2]8BQF_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:3]4X8M_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:4]6S36_A.pdb    KIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:5]9R6U_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:6]9R71_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:7]8Q2B_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKA
[Truncated_Name:8]8RJ9_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:9]6RZE_A.pdb    AIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:10]4X8H_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
```

```
[Truncated_Name:11]3HPR_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDGTGEELTTRKD
[Truncated_Name:12]1E4V_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:13]5EJE_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:14]1E4Y_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:15]3X2S_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:16]6HAP_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:17]6HAM_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:18]8PVW_A.pdb   RILKR--GETSGRV------------------------D
[Truncated_Name:19]4K46_A.pdb   RMAGRRAHLASGRTYHNVYNPPKVEGKDDVTGEDLVIRED
[Truncated_Name:20]4NP6_A.pdb   RMAGRRAHLPSGRTYHVVYNPPKVEGKDDVTGEDLVIRED
                                 ^  *     ***
                              121        .         .         .         160

                              161        .         .         .         200
[Truncated_Name:1]1AKE_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:2]8BQF_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:3]4X8M_A.pdb    DQEETVRKRLVEWHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:4]6S36_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:5]9R6U_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:6]9R71_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:7]8Q2B_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:8]8RJ9_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:9]6RZE_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:10]4X8H_A.pdb   DQEETVRKRLVEYHQMTAALIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:11]3HPR_A.pdb   DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:12]1E4V_A.pdb   DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:13]5EJE_A.pdb   DQEECVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:14]1E4Y_A.pdb   DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:15]3X2S_A.pdb   DQEETVRKRLCEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:16]6HAP_A.pdb   DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:17]6HAM_A.pdb   DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:18]8PVW_A.pdb   DNEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:19]4K46_A.pdb   DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG
[Truncated_Name:20]4NP6_A.pdb   DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG
                                * ** *  **  ^*  ** ** ** *** ** * * * **
                              161        .         .         .         200

                              201        .     216
[Truncated_Name:1]1AKE_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:2]8BQF_A.pdb    TKPVAEVRADLEKIL-
[Truncated_Name:3]4X8M_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:4]6S36_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:5]9R6U_A.pdb    TKPVAEVRADLEKILG
```

```
[Truncated_Name:6]9R71_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:7]8Q2B_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:8]8RJ9_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:9]6RZE_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:10]4X8H_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:11]3HPR_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:12]1E4V_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:13]5EJE_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:14]1E4Y_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:15]3X2S_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:16]6HAP_A.pdb   TKPVCEVRADLEKILG
[Truncated_Name:17]6HAM_A.pdb   TKPVCEVRADLEKILG
[Truncated_Name:18]8PVW_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:19]4K46_A.pdb   TKAVAEVSAELEKALA
[Truncated_Name:20]4NP6_A.pdb   TKQVSEVSADIAKALA
                                ** * ** *^^ * *
                              201         .     216

Call:
  pdbaln(files = files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  20 sequence rows; 216 position columns (182 non-gap, 34 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

Quick interactive structure view

```r
#view.pdbs(pdbs)
```

```r
# Vector containing PDB database codes
ids <- basename.pdb(pdbs$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
```

```
[3] "Escherichia coli O139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Vibrio cholerae O1 biovar El Tor str. N16961"
```

```
anno
```

```
       structureId chainId macromoleculeType chainLength experimentalTechnique
1AKE_A        1AKE       A           Protein         214                 X-
ray
8BQF_A        8BQF       A           Protein         234                 X-
ray
4X8M_A        4X8M       A           Protein         214                 X-
ray
6S36_A        6S36       A           Protein         214                 X-
ray
9R6U_A        9R6U       A           Protein         214                 X-
ray
9R71_A        9R71       A           Protein         214                 X-
ray
8Q2B_A        8Q2B       A           Protein         214                 X-
ray
8RJ9_A        8RJ9       A           Protein         214                 X-
ray
6RZE_A        6RZE       A           Protein         214                 X-
ray
4X8H_A        4X8H       A           Protein         214                 X-
ray
3HPR_A        3HPR       A           Protein         214                 X-
ray
1E4V_A        1E4V       A           Protein         214                 X-
ray
5EJE_A        5EJE       A           Protein         214                 X-
ray
1E4Y_A        1E4Y       A           Protein         214                 X-
ray
3X2S_A        3X2S       A           Protein         214                 X-
ray
6HAP_A        6HAP       A           Protein         214                 X-
ray
6HAM_A        6HAM       A           Protein         214                 X-
ray
```

```
8PVW_A         8PVW       A           Protein         187                  X-
ray
4K46_A         4K46       A           Protein         214                  X-
ray
4NP6_A         4NP6       A           Protein         217                  X-
ray
       resolution       scopDomain                                        pfam
1AKE_A      2.000 Adenylate kinase                      Adenylate kinase (ADK)
8BQF_A      2.050             <NA>                      Adenylate kinase (ADK)
4X8M_A      2.600             <NA>                      Adenylate kinase (ADK)
6S36_A      1.600             <NA>                      Adenylate kinase (ADK)
9R6U_A      1.770             <NA>                      Adenylate kinase (ADK)
9R71_A      1.610             <NA>                      Adenylate kinase (ADK)
8Q2B_A      1.760             <NA> Adenylate kinase, active site lid (ADK_lid)
8RJ9_A      1.590             <NA> Adenylate kinase, active site lid (ADK_lid)
6RZE_A      1.690             <NA>                      Adenylate kinase (ADK)
4X8H_A      2.500             <NA>                      Adenylate kinase (ADK)
3HPR_A      2.000             <NA>                      Adenylate kinase (ADK)
1E4V_A      1.850 Adenylate kinase                      Adenylate kinase (ADK)
5EJE_A      1.900             <NA>                      Adenylate kinase (ADK)
1E4Y_A      1.850 Adenylate kinase                      Adenylate kinase (ADK)
3X2S_A      2.800             <NA>                                        <NA>
6HAP_A      2.700             <NA>                      Adenylate kinase (ADK)
6HAM_A      2.550             <NA>                      Adenylate kinase (ADK)
8PVW_A      2.000             <NA>                      Adenylate kinase (ADK)
4K46_A      2.010             <NA>                      Adenylate kinase (ADK)
4NP6_A      2.004             <NA>                      Adenylate kinase (ADK)
              ligandId
1AKE_A             AP5
8BQF_A             AP5
4X8M_A            <NA>
6S36_A CL (3),NA,MG (2)
9R6U_A       AP5,NA,GOL
9R71_A             AP5
8Q2B_A      SO4,MPO,AP5
8RJ9_A         ADP (2)
6RZE_A   NA (3),CL (2)
4X8H_A            <NA>
3HPR_A             AP5
1E4V_A             AP5
5EJE_A          AP5,CO
1E4Y_A             AP5
3X2S_A  JPY (2),AP5,MG
```

```
6HAP_A              AP5
6HAM_A              AP5
8PVW_A              AP5
4K46_A      ADP,AMP,PO4
4NP6_A             <NA>
                                                                               ligandName
1AKE_A                                                   BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
8BQF_A                                                   BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
4X8M_A                                                                               <NA>
6S36_A                                      CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
9R6U_A                               BIS(ADENOSINE)-5'-PENTAPHOSPHATE,SODIUM ION,GLYCEROL
9R71_A                                                   BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
8Q2B_A SULFATE ION,3[N-MORPHOLINO]PROPANE SULFONIC ACID,BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
8RJ9_A                                                       ADENOSINE-5'-
DIPHOSPHATE (2)
6RZE_A                                                    SODIUM ION (3),CHLORIDE ION (2)
4X8H_A                                                                               <NA>
3HPR_A                                                   BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
1E4V_A                                                   BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
5EJE_A                                   BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A                                                   BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
3X2S_A  N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A                                                   BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
6HAM_A                                                   BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
8PVW_A                                                   BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
4K46_A                    ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION
4NP6_A                                                                               <NA>
                                                  source
1AKE_A                              Escherichia coli
8BQF_A                              Escherichia coli
4X8M_A                              Escherichia coli
6S36_A                              Escherichia coli
9R6U_A                              Escherichia coli
```

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9R71_A                            Escherichia coli
8Q2B_A                            Escherichia coli
8RJ9_A                            Escherichia coli
6RZE_A                            Escherichia coli
4X8H_A                            Escherichia coli
3HPR_A                       Escherichia coli K-12
1E4V_A                            Escherichia coli
5EJE_A       Escherichia coli O139:H28 str. E24377A
1E4Y_A                            Escherichia coli
3X2S_A     Escherichia coli str. K-12 substr. MDS42
6HAP_A       Escherichia coli O139:H28 str. E24377A
6HAM_A                       Escherichia coli K-12
8PVW_A                       Escherichia coli K-12
4K46_A                  Photobacterium profundum
4NP6_A Vibrio cholerae O1 biovar El Tor str. N16961

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBI
8BQF_A
4X8M_A
6S36_A
9R6U_A                                                                                     C
9R71_A                                                                                   Crys
8Q2B_A                                                 E. coli Adenylate Kinase variant D158A (
8RJ9_A                                                           E. coli adenylate kinase Asp84
6RZE_A
4X8H_A
3HPR_A
1E4V_A
loop
5EJE_A                                                                                   Cryst
1E4Y_A
loop
3X2S_A
conjugated adenylate kinase
6HAP_A
6HAM_A
8PVW_A
4K46_A
4NP6_A
                                                       citation rObserved   rFree
1AKE_A             Muller, C.W., et al. J Mol Biology (1992)   0.19600      NA
8BQF_A  Scheerer, D., et al. Proc Natl Acad Sci U S A (2023)   0.22073 0.25789
4X8M_A               Kovermann, M., et al. Nat Commun (2015)   0.24910 0.30890
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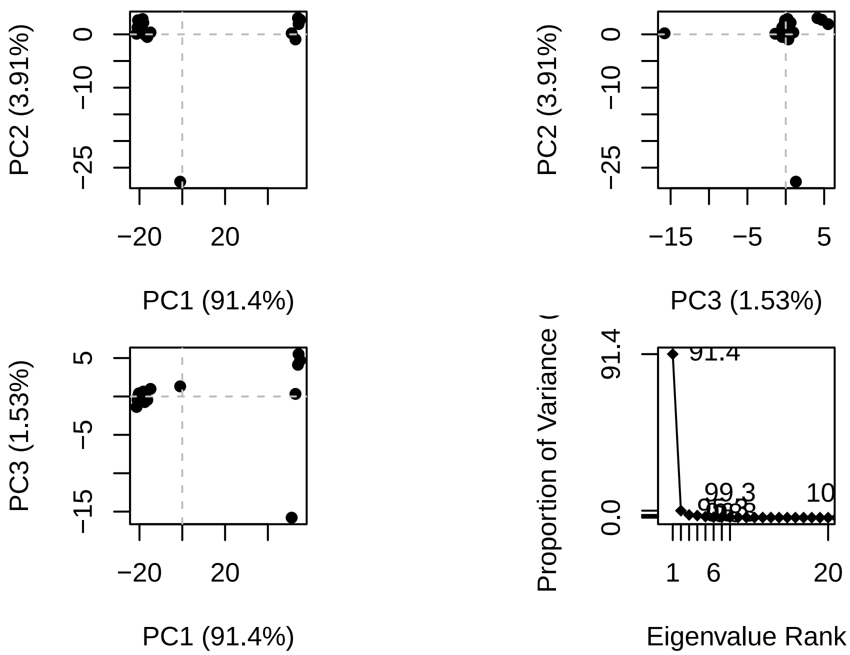
6S36_A                 Rogne, P., et al. Biochemistry (2019)   0.16320 0.23560
9R6U_A              Mattsson, J., et al. Biochemistry (2025)        NA 0.22790
9R71_A              Mattsson, J., et al. Biochemistry (2025)   0.19600 0.24400
8Q2B_A               Nam, K., et al. J Chem Inf Model (2024)   0.18320 0.22440
8RJ9_A                        Nam, K., et al. Sci Adv (2024)   0.15190 0.20290
6RZE_A                 Rogne, P., et al. Biochemistry (2019)   0.18650 0.23500
4X8H_A               Kovermann, M., et al. Nat Commun (2015)   0.19610 0.28950
3HPR_A Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)   0.21000 0.24320
1E4V_A                  Muller, C.W., et al. Proteins (1993)   0.19600      NA
5EJE_A Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)   0.18890 0.23580
1E4Y_A                  Muller, C.W., et al. Proteins (1993)   0.17800      NA
3X2S_A               Fujii, A., et al. Bioconjug Chem (2015)   0.20700 0.25600
6HAP_A              Kantaev, R., et al. J Phys Chem B (2018)   0.22630 0.27760
6HAM_A              Kantaev, R., et al. J Phys Chem B (2018)   0.20511 0.24325
8PVW_A              Rodriguez, J.A., et al. To be published    0.18590 0.23440
4K46_A                   Cho, Y.-J., et al. To be published    0.17000 0.22290
4NP6_A                      Kim, Y., et al. To be published    0.18800 0.22200
         rWork spaceGroup
1AKE_A 0.19600  P 21 2 21
8BQF_A 0.21882  P 2 21 21
4X8M_A 0.24630    C 1 2 1
6S36_A 0.15940    C 1 2 1
9R6U_A 0.19190  P 21 2 21
9R71_A 0.19300  P 21 21 2
8Q2B_A 0.18100   P 1 21 1
8RJ9_A 0.15010  P 21 21 2
6RZE_A 0.18190    C 1 2 1
4X8H_A 0.19140    C 1 2 1
3HPR_A 0.20620  P 21 21 2
1E4V_A 0.19600  P 21 2 21
5EJE_A 0.18630  P 21 2 21
1E4Y_A 0.17800   P 1 21 1
3X2S_A 0.20700 P 21 21 21
6HAP_A 0.22370    I 2 2 2
6HAM_A 0.20311       P 43
8PVW_A 0.18340  P 2 21 21
4K46_A 0.16730 P 21 21 21
4NP6_A 0.18600       P 43
```

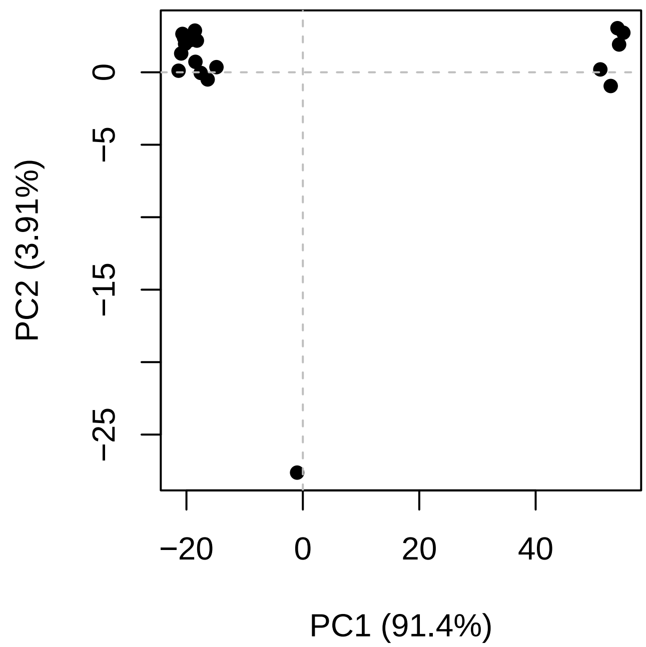
Principal Component Analysis

PCA of all this structural data (x, y and z atom coordinates):

```
# Perform PCA
pc <- pca(pdbs)
plot(pc)
```




```
# Create plot with PC 1 and 2
plot(pc, 1:2)
```



Interactive view of the PC1 captured structural differences:

```
# Create interactive view of the PC1 captured structural differences

#| eval: !expr knitr::is_html_output()

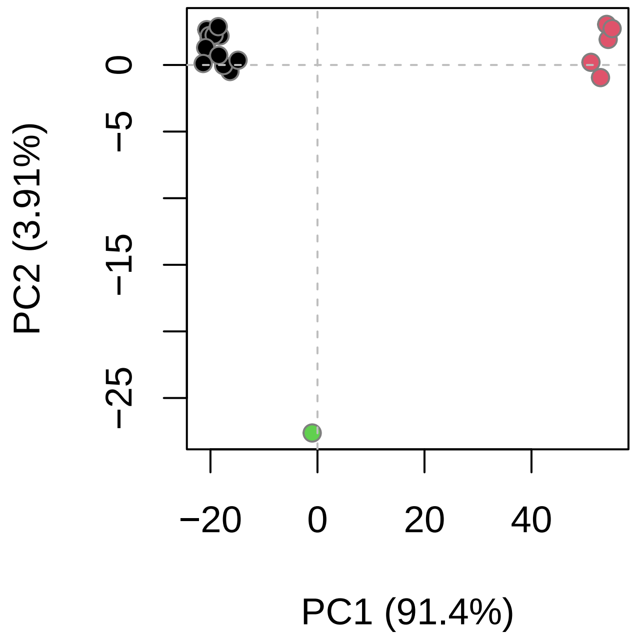
#view.pca(pc)
```

```
# Calculate RMSD
rd <- rmsd(pdbs)
```

```
Warning in rmsd(pdbs): No indices provided, using the 182 non NA positions
```

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```




```r
# Visualize first principal component
pc1 <- mktrj(pc, pc=1, file="pca.pdb")
```